

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457761.8549 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.024
Transit depth (Rp/Rs)^2	3.73 +/- 0.94 [%]
Orbital Inclination (inc)	80.34 +/- 1.5
Airmass coefficient 1 (a1)	0.983 +/- 0.02
Airmass coefficient 2 (a2)	0.0094 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	2.01 %
Best Comparison Star	#3 - [48, 259]
Optimal Aperture	2.91
Optimal Annulus	15.74
Transit Duration (day)	0.0464 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.193 $\pm$ 0.024 vs 0.1594 (deviation 1.4 σ)
Duration fit vs published	0.0464 d vs 0.0512 d (deviation 0.68 σ)
Mid-time O-C	-0.2 min
Depth SNR	8.0
Residual scatter	2.01 %

Light curve

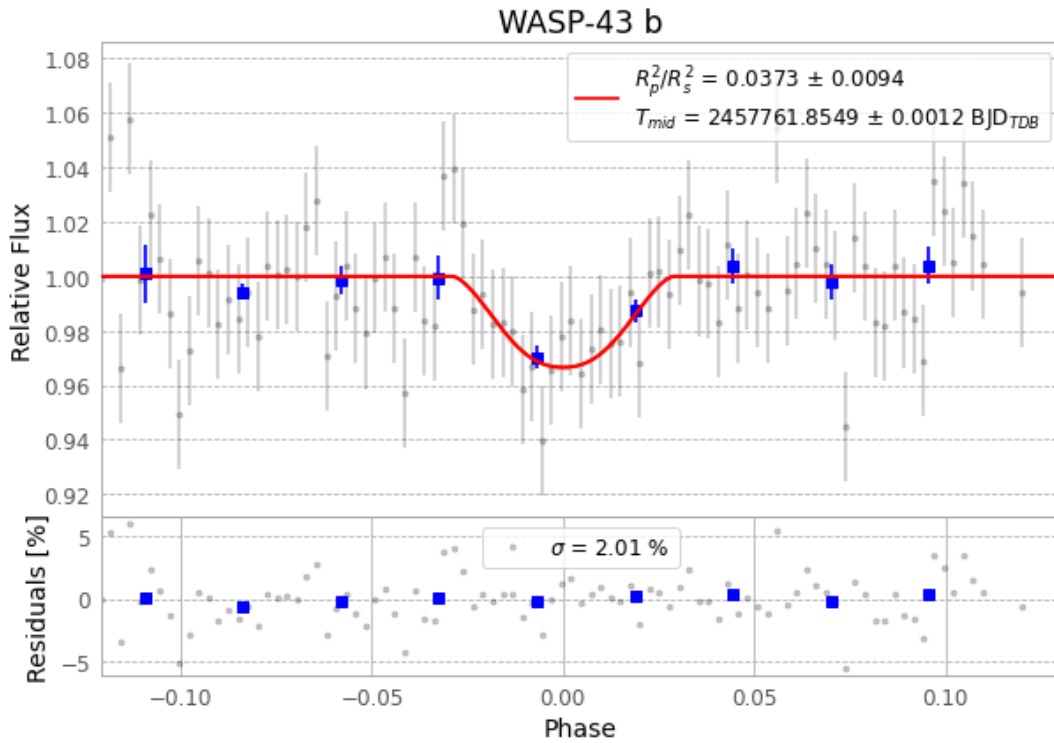


Figure 1 — FinalLightCurve_WASP-43 b_2017-01-07.png (SHA-256 65671f1c88f4e109...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [179, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5dfcd84a6b1d4f1d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

103 FITS frames (68 MB), WASP-43170108060312.FITS → WASP-43170108110912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170108.log (b5f18e3c66be...), FinalLightCurve_WASP-43 b_2017-01-07.png (65671f1c88f4...), FinalLightCurve_WASP-43 b_2017-01-07.csv (5056613dfb6c...)

Compute resources

Wall time 445.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-26 00:20Z.